Topic 7: Performance measurement

Sessions 18-19-20

Accounting-based measures of performance

4 Key Performance Indicators (KPI)

- ✓Return on investment (ROI)
- ✓Residual income (RI)
- ✓Economic value added (EVA)
- ✓Return on sales (ROS)

ROI

$$ROI = \frac{Income}{Investment}$$

- Operating income
- Net income

- Total assets
- Assets – Current Liabilities

ROI

Income statement				
	San Francisco Hotel	**Chicago Hotel**	**New Orleans Hotel**	**Total**
Hotel revenues	1.200.000	1.400.000	3.185.000	5.785.000
Hotel variable costs	310.000	375.000	995.000	1.680.000
Hotel fixed costs	650.000	725.000	1.680.000	3.055.000
HOTEL OPERATING INCOME	**240.000**	**300.000**	**510.000**	**1.050.000**
Interest costs LT debt				450.000
Income before taxes				600.000
Income taxes (30%)				180.000
NET INCOME				**420.000**

ROI

Balance sheet (end 2012)				
	San Francisco Hotel	**Chicago Hotel**	**New Orleans Hotel**	**Total**
Current assets	400.000	500.000	660.000	1.560.000
Long-term assets	600.000	1.500.000	2.340.000	4.440.000
TOTAL ASSETS	**1.000.000**	**2.000.000**	**3.000.000**	**6.000.000**
Current liabilities	50.000	150.000	300.000	500.000
Long-term debt				4.500.000
Stockholders' equity				1.000.000
TOTAL LIABILITIES & EQUITY				**6.000.000**

ROI

ROI	San Francisco Hotel	Chicago Hotel	New Orleans Hotel	Total
Total Assets	1.000.000	2.000.000	3.000.000	6.000.000
Operating Income	240.000	300.000	510.000	1.050.000
ROI	**24,0%**	**15,0%**	**17,0%**	**17,5%**
Total Assets - Current Liabilities				5.500.000
Net Income				420.000
ROI				**7,6%**

ROI

Companies should pay attention not only to increasing income but also to reduce investment:

- ✓ Decrease idle cash
- ✓ Manage credit judiciously
- ✓ Proper inventory levels (working capital management)
- ✓ Spending carefully on long-term assets

Dupont method

$$\text{ROI} = \frac{Income}{Investment} = \frac{Income}{Revenues} * \frac{Revenues}{Investment}$$

Return on sales

Investment turnover

Dupont method

ROI	San Francisco Hotel	Chicago Hotel	New Orleans Hotel
Operating Income	240.000	300.000	510.000
Hotel Revenues	1.200.000	1.400.000	3.185.000
Total Assets	1.000.000	2.000.000	3.000.000
Income / revenues (ROS)	20,0%	21,4%	16,0%
Revenues / Investments (Inv. Turnover)	120,0%	70,0%	106,2%
ROI (Op. Income / Investments)	**24,0%**	**15,0%**	**17,0%**

To increase ROI:

- ✓Decrease investment
- ✓Increase revenues
- ✓Decrease costs

→ ✓Improve Op. Income

ROI: versatile / simple / used

- Financial investment
- Marketing expenses
- Business unit
- Real state investment
- …

LIMITATIONS:

- Time is not considered
- Encompasses different measures

Residual Income (RI)

$$Residual\ income$$

$$= Income\ - Required\ rate\ of\ return\ * Investment$$

The return earned in excess of the required return for investments

Residual Income (RI)

Residual Income	San Francisco Hotel	Chicago Hotel	New Orleans Hotel
Total Assets	1.000.000	2.000.000	3.000.000
Operating Income	240.000	300.000	510.000
Required Return	**12,0%**	**12,0%**	**12,0%**
Required Op. Income	120.000	240.000	360.000
Residual Income	**120.000**	**60.000**	**150.000**
RI / Total Assets	12,0%	3,0%	5,0%
ROI = (RI/T. Assets) + Required Return	24,0%	15,0%	17,0%

Residual Income (RI)

- ✓ E.g., San Francisco Hotel upgrades room furnishing and other features, the operating income will increase 70,000 and the investment (assets) 400,000
- ✓ ROI (of the expansion) = 70,000 / 400,000 = 17.5%
- ✓ Required return on an investment = 12%

Residual income (RI)

Example (assumptions)	**Post-Invest.**	**Pre-Invest.**	
San Francisco upgrades rooms; capex	400.000		
Operating income expected to increase by $	70.000		
ROI of expansion	**17,5%**		
Required Return	12,0%	12,0%	**Change**
Total Assets post-investment	1.400.000	1.000.000	**400.000**
Operating income post-investment	310.000	240.000	**70.000**
ROI	**22,1%**	**24,0%**	**-1,9%**
Required Return $	168.000	120.000	**48.000**
Residual Income	**142.000**	**120.000**	**22.000**

Would you recommend this capex? Why?

Economic Value Added

EVA = After-tax operating income − [Weighted average cost of capital X (Assets – current liabilities)]

WACC

WACC

$$\text{WACC} = \frac{\text{(after-tax cost of debt * market value of LT debt)} + \text{(cost of equity * market value of equity)}}{\text{(market value of LT debt + market value of equity)}}$$

Weighted average cost of co's external funding (i.e. Banks + Shareholders)

EVA

- ✓ After-tax cost of debt = 0,1 x (1 – 0.3) = 0.07 (7%)
- ✓ Market value of long-term debt = 4,500,000

  (rule of thumb: loans = face value ; bonds = market value)

- ✓ Cost of equity = 0.14 (14%)
- ✓ Market value of equity = 4,500,000

$$\text{WACC} = \frac{(0.07 \times 4{,}500{,}000) + (0.14 \times 4{,}500{,}000)}{(4{,}500{,}000 + 4{,}500{,}000)} = 0.105\ (10{,}5\%)$$

EVA

Hotel	After-tax operating income		WACC	Total assets	Current liabilities	EVA
San Francisco	=240,000*0.7	168,000	0.105	1,000,000	50,000	68,250
Chicago	=300,000*0.7	210,000	0.105	2,000,000	150,000	15,750
New Orleans	=510,000*0.7	357,000	0.105	3,000,000	300,000	73,500

EVA

EVA = PAT - (WACC x (Assets - Curr. Liab.))	**San Francisco Hotel**	**Chicago Hotel**	**New Orleans Hotel**	**Total**
TOTAL ASSETS	1.000.000	2.000.000	3.000.000	6.000.000
Current liabilities	50.000	150.000	300.000	500.000
HOTEL OPERATING INCOME	240.000	300.000	510.000	1.050.000
Market value of LTD				4.500.000
Book Value of Equity				1.000.000
Market value of Equity				4.500.000
Tax rate				30,0%
Cost of Debt				10,0%
Cost of Debt (after tax)				7,0%
Cost of Equity				14,0%
WACC	10,5%	10,5%	10,5%	**10,5%**
After tax Operating Income	168.000	210.000	357.000	735.000
Total Assets - Current Liabilities	950.000	1.850.000	2.700.000	5.500.000
Required return based on WACC	99.750	194.250	283.500	577.500
EVA	**68.250**	**15.750**	**73.500**	**157.500**

Return on Sales

$$\text{ROS} = \frac{\text{Income}}{\text{Revenues}}$$

Return on Sales

Hotel	Operating income	Revenues	ROS
San Francisco	240,000	1,200,000	20%
Chicago	300,000	1,400,000	21%
New Orleans	510,000	3,185,000	16%

Investment definitions and asset measurement to calculate ROI at Current Values

Alternative investment definitions

- ✓ Total assets available
- ✓ Total assets employed
- ✓ Total assets employed - current liabilities
- ✓ Equity

Alternative asset measurement

✓ *Historical cost or current cost?*

✓ *Gross value or net (book) value?*

Current cost: cost of purchasing an asset (today) identical to the one currently held or that provides services like the one currently held

Net assets at historical cost → Net assets at current cost

Formulas

$$Yearly\ depreciation\ = \frac{Gross\ value}{Useful\ life}$$

$$Net\ value = Gross\ value\ - Age\ of\ facility\ * Yearly\ depreciation$$

Net LT assets (historical) to Gross LT assets (historical)

From NET to GROSS LT Assets	San Francisco Hotel	Chicago Hotel	New Orleans Hotel
Net LT Assets (end 2012)	600.000	1.500.000	2.340.000
Age of facility in years (end 2012)	8	4	2
Year of construction	2.004	2.008	2.010
Useful life (years for depreciation)	14	14	14
Remainig useful life	6	10	12
Yearly depreciation	**100.000**	**150.000**	**195.000**
Accum. Depreciation	800.000	600.000	390.000
Gross LT Assets (Historical)	**1.400.000**	**2.100.000**	**2.730.000**

Gross LT assets (historical) to Gross LT assets (current)

Year	Constr. Cost index	Coefficient to Current
2.004	**100**	**1,800**
2.005	110	1,636
2.006	122	1,475
2.007	136	1,324
2.008	**144**	**1,250**
2.009	152	1,184
2.010	**160**	**1,125**
2.011	174	1,034
2.012	180	1,000

Gross LT assets (historical) to Gross LT assets (current)

	GROSS LT Assets: from historical to current	**San Francisco Hotel**	**Chicago Hotel**	**New Orleans Hotel**
1	Gross LT Assets (Historical)	1.400.000	2.100.000	2.730.000
2	Year of construction	2.004	2.008	2.010
3	Construction index 2012	180	180	180
4	Construction index year of construction	100	144	160
5	Times revaluated (3/4)	**1,800**	**1,250**	**1,125**
6	**Gross LT Assets (Curr. Val.) (6=1x5)**	**2.520.000**	**2.625.000**	**3.071.250**

Gross LT assets (current) to Net LT assets (current)

From GROSS to NET LT Assets (Curr.)	San Francisco Hotel	Chicago Hotel	New Orleans Hotel
Gross LT Assets (Current value)	2.520.000	2.625.000	3.071.250
Yearly Depreciation on Current Value	180.000	187.500	219.375
Accum. depreciation on Current Value	1.440.000	750.000	438.750
Net LT Assets (Current value)	**1.080.000**	**1.875.000**	**2.632.500**

Net total assets (current value)

NET TOTAL ASSETS (current value)	**San Francisco Hotel**	**Chicago Hotel**	**New Orleans Hotel**
Current Assets	400.000	500.000	660.000
Net LT Assets Current	1.080.000	1.875.000	2.632.500
Total Assets Current	**1.480.000**	**2.375.000**	**3.292.500**

Operating income (using current cost depreciation)

Operating Income using current cost depreciation	**San Francisco Hotel**	**Chicago Hotel**	**New Orleans Hotel**
Operating Income	240.000	300.000	510.000
Yearly Depreciation (book value)	100.000	150.000	195.000
Yearly Depreciation (current value)	180.000	187.500	219.375
Difference	80.000	37.500	24.375
Oper. Income (current cost depr.)	**160.000**	**262.500**	**485.625**

ROI at current cost

ROI using current cost	**San Francisco Hotel**	**Chicago Hotel**	**New Orleans Hotel**
Operating Income (current cost depr.)	160.000	262.500	485.625
Total Assets (net at current cost)	1.480.000	2.375.000	3.292.500
ROI at Current Cost	**10,8%**	**11,1%**	**14,7%**
ROI at Historical Cost	24,0%	15,0%	17,0%

References

- Horngren et al., 2015: chapter 23 (chapter 24 in 17th ed.)